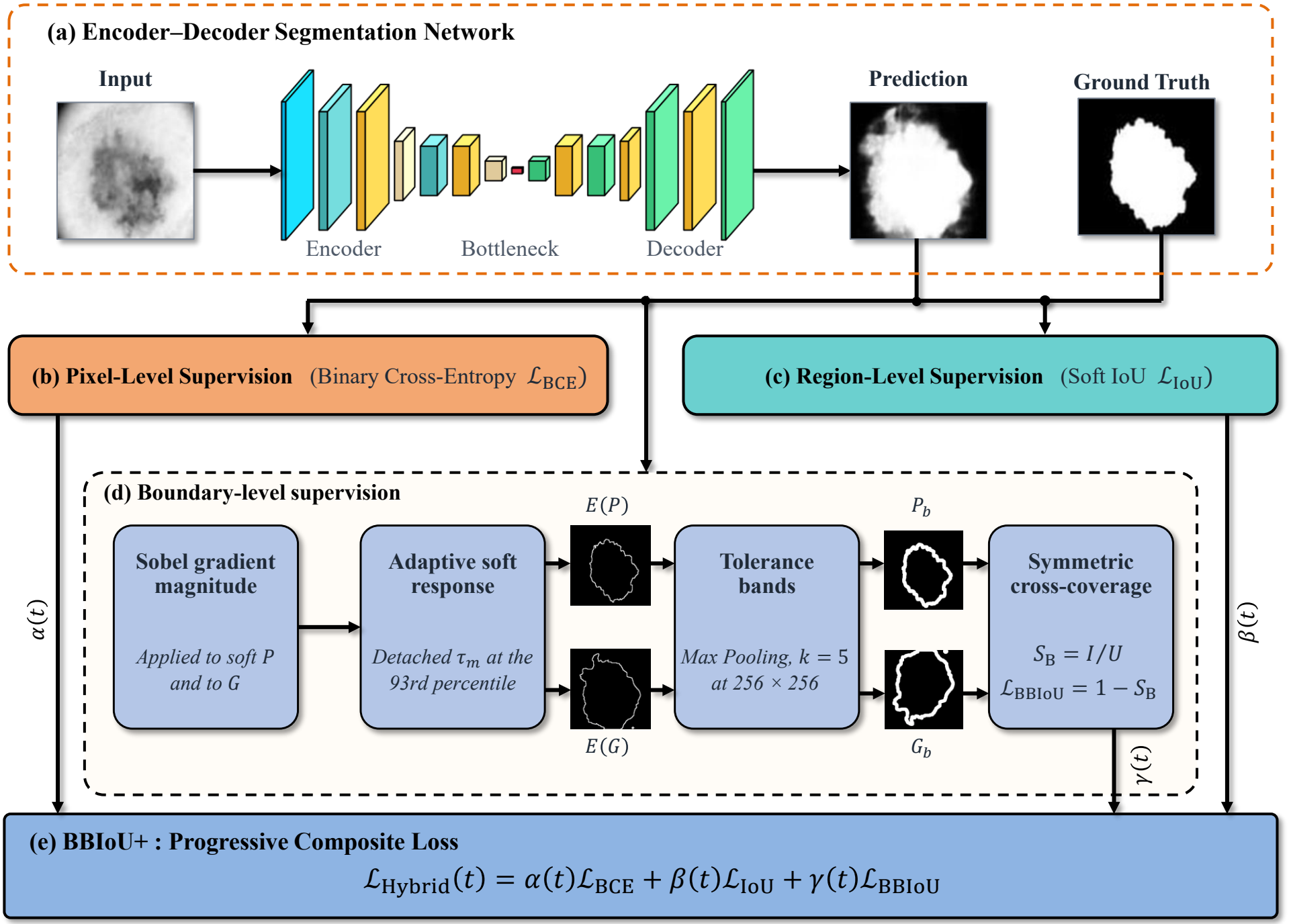

(a) Encoder–Decoder Segmentation Network
Input
Encoder
Bottleneck
Decoder
Prediction
Ground Truth
(b) Pixel-Level Supervision (Binary Cross-Entropy $\mathcal{L}_{\mathrm{BCE}}$)
(c) Region-Level Supervision (Soft IoU $\mathcal{L}_{\mathrm{IoU}}$)
(d) Boundary-level supervision
Sobel gradient magnitude
Applied to soft P and to G
Adaptive soft response
Detached τ_m at the 93rd percentile
$E(P)$
$E(G)$
Tolerance bands
Max Pooling, $k = 5$ at 256 × 256
P_b
G_b
Symmetric cross-coverage
$S_{\mathrm{B}} = I/U$
$\mathcal{L}_{\mathrm{BBIoU}} = 1 - S_{\mathrm{B}}$
$\alpha(t)$
$\beta(t)$
$\gamma(t)$
(e) BBIoU+ : Progressive Composite Loss
$\mathcal{L}_{\mathrm{Hybrid}}(t) = \alpha(t)\mathcal{L}_{\mathrm{BCE}} + \beta(t)\mathcal{L}_{\mathrm{IoU}} + \gamma(t)\mathcal{L}_{\mathrm{BBIoU}}$